Summary in Graph

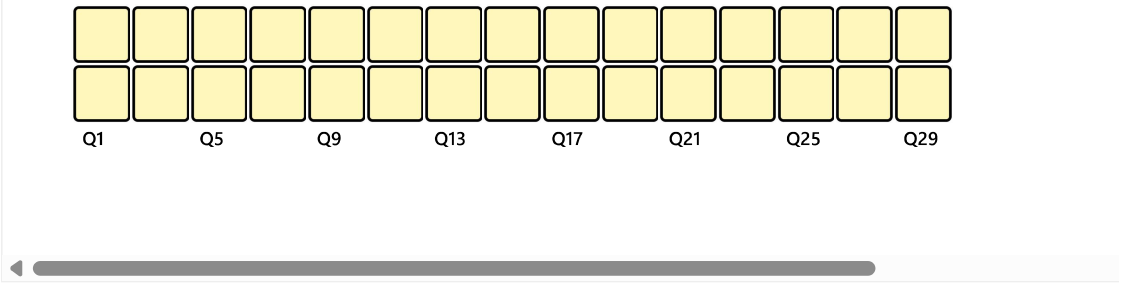
Exam Summary (GO Classes DA Test Series 2026 | Data Structures & Algorithms | Subject Wise Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (20 + 10)
Total Marks:	40 (20 + 20)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (25) out of 253 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following code fragment written in Python:

```
    def bar(n):
        if n <= 10:
            return 0

5.      if n < 100:
            r = 0
            for i in range(0, n, 1):   # loop runs n times
                r += 1
            # note: no return here
10.
        return bar(n/3) + bar(n-1)
```

What will be the recurrence relation for the running time complexity of function `bar(n)`, where n is a power of 2?

A. $$T(n) = \begin{cases} 0 & n \leq 10 \\ c_2 n & n < 100 \\ T\left(\frac{n}{3}\right) + T(n-1) + c_3 & \text{otherwise} \end{cases}$$

B. $$T(n) = \begin{cases} 0 & n \leq 10 \\ T\left(\frac{n}{3}\right) + T(n-1) + c_2 n + c_3 & n < 100 \end{cases}$$

C. $$T(n) = \begin{cases} 0 & n \leq 10 \\ T\left(\frac{n}{3}\right) + T(n-1) + c_2 n + c_3 & \text{otherwise} \end{cases}$$

D. $$T(n) = \begin{cases} 0 & n \leq 10 \\ c_2 \log_3(n) & n < 100 \\ T\left(\frac{n}{3}\right) + T(n-1) + c_3 & \text{otherwise} \end{cases}$$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Consider the postorder traversal of a Complete Binary Tree given as follows:

$$65, 71, 12, 43, 90, 22, 33, 51$$

Which of the following is the correct preorder traversal?

A. $51, 65, 71, 12, 43, 22, 90, 33$

B. $51, 12, 65, 71, 33, 22, 43, 90$

C. $51, 43, 71, 65, 12, 33, 90, 22$

D. $51, 65, 71, 33, 43, 90, 12, 22$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

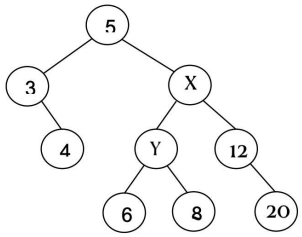
An graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ is an undirected simple graph of arbitrary number of edges E and having $|\mathrm{V}| = 10$ nodes. The diameter *d* of G is defined as

$$\text{diameter}(G) = \max_{u,v \in V} \delta(u, v)$$

such $\delta(u, v)$ is the shortest distance between vertices u and v. What is the maximum possible value of d is _____.

Your Answer: | Correct Answer: 9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures



The graph shown above is a binary search tree of distinct positive integers. What will be the minimum possible difference of unknown nodes X to Y ?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Suppose you combine open addressing with a limited form of chaining. You build an table with m slots that can store two keys in each slot.
1. Hash (key, probe number) to one of the m slots.
2. If the slot has less than two keys, insert it there.
3. Otherwise, increment the probe number and go to step 1.
Assume that there are exactly k slots in the table that are completely full. What is the probability that the first probe is successful?

A. $(\mathrm{m} - \mathrm{km})/\mathrm{d}$
B. k/m
C. $(\mathrm{m} - 2\mathrm{k})/2\ \mathrm{m}$
D. $(\mathrm{m} - \mathrm{k})/\mathrm{m}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

A linear-probing hash table of length 10 uses the hash function $\mathbf{h}(\mathbf{k}) = \mathbf{k}$ mod 10 to hash integer keys between 0 and 100. After inserting 8 keys into an empty hash table, the table is as shown below.

index	0	1	2	3	4	5	6	7	8	9
value	20	19	–	–	14	44	6	74	86	59

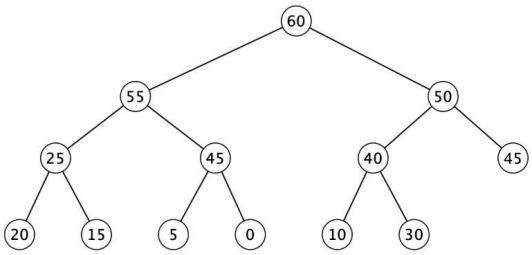
Which of the following choices gives a possible order in which the keys could have been inserted into an initially empty hash table? Assume that the initial length of the array is 10 and that it neither grows nor shrinks.

A. $14, 44, 6, 74, 86, 20, 59, 19$

B. $59, 6, 14, 20, 44, 19, 74, 86$
C. $20, 59, 6, 74, 14, 86, 44, 19$
D. $6, 59, 14, 20, 19, 44, 86, 74$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures



Suppose that the last operation in the binary heap was a call to insert(). Which key/s could have been the last one inserted?

A. 30
B. 40
C. 50
D. 60

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

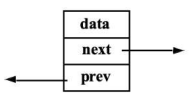
Suppose a binary search tree with 1000 distinct elements is also a complete binary tree. The tree is stored using the array representation of binary heap trees. Assuming that the array indices start with 0, the 2nd largest element of the tree is stored at Index .

A. 254
B. 255
C. 510
D. 511

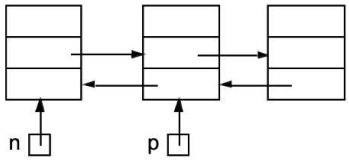
Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Nodes for a doubly linked list are defined to have the following structure:



The next instance variable stores a reference to the next node in the list, and the prev instance variable refers to the previous node in the list.
Below is a list of three of these nodes, along with two reference variables, n and p, that refer to specific nodes in the list.



Which of the following expressions does not refer to the third node in the list?

A. p.next
B. n.next.next
C. p.prev.next
D. n.next.next.prev.next

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Let L be a singly-linked list and X and Y be additional pointer variables such that X points to the first node of L and Y points to the last node of L. Which of the following operations cannot be done in time that is bounded above by a constant?

A. Delete the first node of L
B. Delete the last node of L
C. Swap first and last node of L
D. Swap the first two nodes of L

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following pseudocode:

```
   procedure MYSTERY(root)
       if root.left is NULL then
           return root
       else
5.         return MYSTERY(root.left)
   end procedure
```

Which of the following statements is/are **True** after the complete execution of the `MYSTERY` function called with the reference to the root of a Binary Search Tree of *n* nodes?

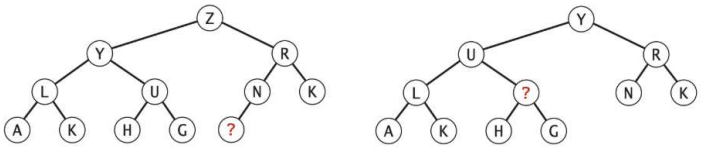
I. It returns the first node visited by an inorder traversal.
II. It returns the last node visited by a postorder traversal.
III. It returns the last node visited by a preorder traversal.

A. I
B. II
C. III
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

A delete operation in the max-heap of distinct keys results in a max-heap in right.



Which of the keys below could be the one labelled with a question mark?

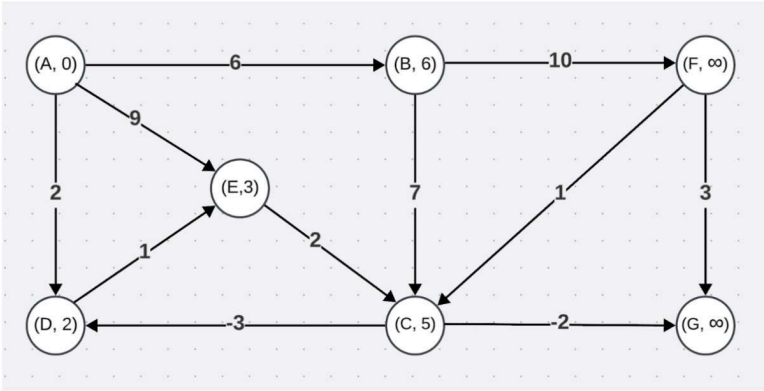
A. I
B. J
C. L
D. M

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Consider the directed weighted graph $\mathrm{G}=(\mathrm{V}, \mathrm{E})$ shown below. Each vertex is represented as (X, d), where X is the node label and d denotes the current shortest distance estimate from the source vertex (A, 0).

At an intermediate stage of Dijkstra's algorithm, after extracting k vertices from the priority queue, the vertex $(\mathrm{k}+1)$ th is extracted. What will be the updated distance estimates of all vertices immediately after relaxing all outgoing edges from the $(\mathrm{k}+1$)th vertex?



A. $(A, 0),(B, 6),(C, 5),(D, 1),(E, 3),(F, \infty),(G, 0)$

B. $(A, 0),(B, 6),(C, 5),(D, 1),(E, 3),(F, \infty),(G, 3)$

C. $(A, 0),(B, 6),(C, 5),(D, 2),(E, 3),(F, \infty),(G, 3)$

D. $(A, 0),(B, 6),(C, 5),(D, 2),(E, 3),(F, \infty),(G, 0)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 05sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following two statements:

- **S1**: If root vertex r has more than one child of a DFS(Depth First Search) tree of an undirected graph G , then removing r will disconnect the graph.

- **S2**: In a BFS(Breadth First Search) tree of a connected undirected graph G, for any vertex v, all its neighbours are at depth at most one.

Which ONE of the following is CORRECT?

A. Both S1 and S2 are TRUE

B. S1 is TRUE and S2 is FALSE

C. S1 is FALSE and S2 is TRUE

D. Both S1 and S2 are FALSE

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider an unordered array of N distinct integers. What is the minimum number of comparisons required to identify all elements except the largest?

A. N - 1

B. N - 1 + $\lceil \log_2 N \rceil$

C. N - 1 + $\lfloor \log_2 N \rfloor$

D. N + $\lceil \log_2 N \rceil - 2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ be an undirected connected graph. Consider the following statements:

- $S1$: The depth of a Breadth-First Search (BFS) tree starting from an arbitrary vertex $\mathrm{v} \in \mathrm{V}$ is equal to the diameter of graph G.
- $S2$: The depth of a Depth-First Search (DFS) tree starting from an arbitrary vertex $v \in V$ is equal to the diameter of graph G.

Which of the above statements is/are TRUE?

A. $S1$ is True and $S2$ is False

B. $S1$ is False and $S2$ is True

C. Both $S1$ and $S2$ are True

D. Both $S1$ and $S2$ are False

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider a binary string S (a string consisting only of 0 s and 1 s) of length n. What is the correct time complexity of sorting the string S?

A. $\Theta(\log n)$

B. $\Theta(n)$

C. $\Theta(n \cdot \log n)$

D. $\Theta\left(n \wedge 2\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following pseudocode:

```
procedure StarDash(n)
  if n = 0 then
     print '*'
   else if n = 1 then
     print '-'
  else
    StarDash(n - 1)
    StarDash(n - 2)
end procedure
```

Which of the following is the correct output for StarDash(5)?

A. - * - - * - **

B. - * - - - * - -

C. - * - - * - * -

D. - * - - * - - -

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider a weighted directed graph $G = (V, E)$ and let X be a shortest path.

- S1: If we double the weight of every edge in the graph, then X will still be the shortest path.
- S2: If we square the weight of every edge in the graph, then X will still be the shortest path.

Which of the above statements are TRUE about X.

A. Both S1 and S2 are True

B. S1 is True and S2 is False

C. S1 is False and S2 is True

D. Both S1 and S2 are False

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ be a simple connected undirected graph with positive edge weights

- $S1$: For such a graph, the Bellman-Ford algorithm and Dijkstra's algorithm always compute the same shortest path weights from a given source but may return different shortest-path trees.
- $S2$: For such a graph, Dijkstra's algorithm always produces the unique shortest-path tree for any arbitrary source.

Which one of the following is CORRECT?

A. Both $S1$ and $S2$ are True

B. $S1$ is True and $S2$ is False

C. $S1$ is False and $S2$ is True

D. Both $S1$ and $S2$ are False

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

A queue is implemented using two stacks, stack1 and stack2. The pseudocode for the implementation is given below:

```
enqueue(x){
    stack1.push(x);
}

dequeue(){
    if( stack2.empty() ){
        while( !stack1.empty() ){
            y = ____(1)____
            _________(2)____
        }
    }
    return stack2.pop();
}
```

Which of the following statements is/are correct?

A. The above two stack queue implementations is incorrect .
B. (1) stack1.pop(); (2) stack2.push(y);
C. (1) stack2.pop(); (2) stack1.push(y);
D. The running time complexity of dequeue operation over a long sequence is $\mathrm{O(n)}$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

You are given a traversal sequence S of distinct positive integers and an unknown tree T.
Which of the following statements is/are TRUE?

A. If T is a binary search tree, then inorder traversal of T can be determined.
B. If T is a complete binary tree and S is preorder traversal, then inorder traversal of T can be determined.

C. If T is a complete binary tree and S is inorder traversal, then preorder traversal of T can be determined.

D. If T is a binary search tree and S is preorder traversal, then postorder traversal of T can be determined.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let A be a one-dimensional array of n distinct real numbers such that $A[i] = f(i)$, where $f : \mathbb{R} \to \mathbb{R}$ is an arbitrary but well-behaved function. It is known that:

- $f(x)$ is continuous and differentiable in the closed interval $[a, b]$,
- $f(x)$ has exactly two real roots in $[a, b]$,
- There exists a unique point $p \in [a, b]$ such that:

$$\lim_{h \to 0^+} f'(p + h) < 0, \quad \lim_{h \to 0^-} f'(p - h) > 0, \quad \text{and } f'(p) = 0$$

Given that the array A is indexed over integers in $[a, b] \cap \mathbb{Z}$, what is the minimum time complexity (in terms of n) required to find the index i closest to the point p where $f(x)$ achieves its maximum value?

A. $\theta(logn)$

B. $\theta(n)$

C. $\theta(1)$

D. $\theta(log^2(n))$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

A system of difference constraints of the form $x_j - x_i \leq b_k$, can be represented by the help of directed graph $\mathrm{G}(\mathrm{V}, \mathrm{E})$, where $V = \{v_0, v_1, \ldots, v_n\}$ and

$$E = \left\{(v_i, v_j) : x_j - x_i \leq b_k\right\} \cup \left\{(v_0, v_1), (v_0, v_2), \ldots, (v_0, v_n)\right\}$$

Find the feasible solution for the following system of difference constraints:

$$\begin{aligned} x_1 - x_2 &\leq 1 \\ x_1 - x_3 &\leq -4 \\ x_2 - x_4 &\leq 7 \\ x_3 - x_2 &\leq 2 \\ x_4 - x_1 &\leq -1 \\ x_4 - x_3 &\leq 3 \end{aligned}$$

A. $x = \{2, 1, 5, 3\}$
B. $x = \{2, 1, 4, 3\}$
C. $x = \{5, 7, 9, 4\}$
D. $x = \{2, 4, 6, 1\}$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Consider a directed graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$, where $\mathrm{V} = \{1, 2, \ldots, 100\}$ and $\mathrm{E} = \{(\mathrm{i}, \mathrm{j}) : \mathrm{i} < \mathrm{j} \leq \mathrm{i} + 2$, for all $\mathrm{i}, \mathrm{j} \in \mathrm{V}\}$. Suppose the adjacency list of each vertex is in decreasing order of vertex number, and depth-first search (DFS) is performed at vertex 1. The number of connected components formed are 'A' and number vertices that will be discovered after vertex 100 is ' B ' then $6\ \mathrm{A} + \mathrm{B}$ _____

Your Answer: | Correct Answer: 55 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

An integer array A is $\boldsymbol{k}$-even-mixed if there are exactly k even integers in A, and the odd integers in A appear in sorted order. Given a k-even-mixed array A containing n distinct integers for $k = \lceil n/\lg n \rceil$, Which of the following is the minimum time complexity to sort A.

A. $\Theta(n)$

B. $\Theta(k.logn)$

C. $\Theta(n.logk)$

D. $\Theta(n.logn)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following program:

```
x := 1
y := 0

  do while y > x:

    x := x/2

    y := y + 1

  end
```

For the above fragment, which of the following is a loop invariant (A loop invariant is a condition that remains true before and after each iteration of the loop)?

A. $x = \frac{1}{2^{y+1}}$

B. $x = \frac{x}{2^y}$

C. $x = \frac{1}{2^y}$

D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider a list of n positive integers a1, a2, . . . , an. Given an integer m, we query to determine if there is a prefix (a1, a2, ... , ai) of the list that sums to m. To solve such a query, we precompute all prefix sums and keep them in an array whose ith element is $\mathrm{a1} + \mathrm{a2} + \ldots +$ ai. The most efficient way of searching this list for integer m is to do a

A. Linear Search

B. Binary Search

C. Random Search (it randomly generates the position of element in O(1) time)

D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following statements about a graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$:

- $S1$: Given a connected unweighted graph G, if a vertex $\mathrm{v} \in \mathrm{V}$ is visited during level k of a Breadth-First Search (BFS) from a source vertex $s \in V$ then every path from s to v has length at least k.
- $S2$: A Depth-First Search (DFS) takes $\Theta\left(V^2\right)$ time on a graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ represented using an adjacency matrix.
- $S3$: Given an adjacency list representation of an undirected graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$, it takes $\mathrm{O}(\mathrm{V})$ time to compute the in-degree of every vertex.

Which of the above statements is/are TRUE?

A. $S1$ is True and $S3$ is True

B. $S1$ is True and $S2$ is False

C. $S1$ is True and $S2$ is True

D. All of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ be an undirected connected graph. Suppose we run a Breadth-First Search (BFS) starting from an arbitrary vertex $v \in V$.

- The depth of the BFS tree is defined as the maximum distance (in number of edges) from v to any other vertex reachable from it. That is:

$$\mathrm{depth}(v) = \max_{u \in V} \delta(v, u)$$

- The diameter of the graph is the maximum shortest-path distance between any two vertices in the graph:

$$\mathrm{diameter}(G) = \max_{s,t \in V} \delta(s, t)$$

Where $\delta(\mathrm{x}, \mathrm{y})$ denotes the shortest-path distance between vertices x and y.
Which of the following statements is correct?

A. The depth of the BFS tree is always equal to the diameter of the graph.
B. The depth of the BFS tree from any vertex is always greater than or equal to the diameter.
C. The depth of the BFS tree from any arbitrary vertex is always less than or equal to the diameter.
D. The depth of the BFS tree from any arbitrary vertex is always equal to the eccentricity(maximum distance from v to any other vertex) of the vertex with maximum degree.

Your Answer: | **Correct Answer: C** | Not Attempted | Time taken: 00min 00sec | Discuss

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